SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

Problem Statement #12: Blood Bank Inventory & Emergency Donor Matcher

Course: Software Engineering Lab 1
Actors: Emergency Requester, Blood Bank Manager, Donor
Status: Approved Baseline (5 FRs & 2 NFRs)

System Scope: Real-time blood bank inventory tracking, component shelf-life monitoring, automated quarantine enforcement, and latency-sensitive geo-targeted emergency donor matching within a 10 km operational perimeter.

REQ ID	TYPE	DESCRIPTION ("THE SYSTEM SHALL...")	PRIORITY	ACCEPTANCE CRITERIA	RATIONALE
FR-001	FUNCTIONAL	The system shall cross-reference emergency blood unit requests against real-time blood bank inventory and broadcast SMS alerts to compatible donors within a 10 km radius.	High	**Pass:** Compatible donors notified via SMS within 30 seconds of emergency flag. **Fail:** Incompatible blood group alerted or latency > 30s.	Minimizes dispatch response time for critical, life-threatening blood shortages.
FR-002	FUNCTIONAL	The system shall automatically flag and quarantine blood bags whose shelf life expires within 24 hours (e.g., platelets after 5 days, RBCs after 42 days).	High	**Pass:** Bag status transitions to "Quarantined" at T−24h and locks from allocation. **Fail:** Expired blood unit remains selectable for dispatch orders.	Prevents transfusion of degraded, ineffective, or hazardous blood components.
FR-003	FUNCTIONAL	The system shall allow the Blood Bank Manager to log new blood donations, update component inventory counts, and record blood types (A/B/AB/O with Rh factor).	High	**Pass:** Inventory records update across DB within 1s of barcode scan. **Fail:** Donation bag logged without mandatory blood type/Rh factor.	Ensures accurate real-time inventory tracking and comprehensive physical stock auditability.
FR-004	FUNCTIONAL	The system shall track and enforce donor eligibility cooldown periods (minimum 56 days for whole blood donations).	Medium	**Pass:** Ineligible donors are suppressed from SMS alert broadcasts until cooldown expires. **Fail:** Donor contacted prior to completing the 56-day cooldown.	Protects voluntary donor health and maintains compliance with standard medical regulations.
FR-005	FUNCTIONAL	The system shall generate a dispatch order and update reserve stock levels immediately upon confirmation of an emergency match.	High	**Pass:** Stock decrements instantly by requested units and outputs a digital dispatch token. **Fail:** Reserved units remain available to other concurrent requesters.	Prevents inventory race conditions and double-allocation during concurrent emergencies.
NFR-001	PERFORMANCE & SECURITY	The inventory ledger must maintain strict transactional consistency (ACID compliance),	High	**Pass:** 0 race conditions / double-allocations under simulated 500 concurrent requests.	High concurrency during multi-casualty disaster events must not corrupt stock integrity.

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		preventing simultaneous allocation of the same blood bag.		**Fail:** Any single blood bag reserved by more than one requester.	
NFR-002	SECURITY & PRIVACY	The system shall encrypt donor personally identifiable information (PII) and geolocation data at rest and in transit using AES-256 and TLS 1.3.	High	**Pass:** Security audit confirms encrypted PII; unauthenticated queries yield HTTP 401/403. **Fail:** Plaintext donor phone numbers/GPS coordinates exposed in logs or payloads.	Complies with statutory healthcare privacy mandates (HIPAA/GDPR) and secures donor confidentiality.

* Formatted in accordance with IEEE 830 / ISO/IEC/IEEE 29148 Software Requirements Engineering Standards.